The University of Melbourne

Semester 2 Examination —November, 2012

Department of Mathematics and Statistics

MAST30027 Modern Applied Statistics

Exam duration: three hours
Reading time: fifteen minutes
This paper has 8 pages.

Authorised materials:
Hand-held electronic calculators may be used.

Instructions to invigilators:
A 16-page script book shall be supplied to each student.

Instructions to students:
There are 10 questions. All questions may be attempted.
The number of marks for each question is indicated after the question.
The total number of marks available is 100.

This paper may be released to the Library

1. Consider i.i.d. random samples $X_1, \cdots, X_n$ of X that has the *pdf*

$$f(x|\theta) = \theta x^{\theta-1}, \quad 0 < x < 1; \quad \theta > 0.$$

Note the following result can be quoted without proof:

(i) The minimum variance bound (i.e. Cramér-Rao lower bound) of an unbiased estimator T of $\tau(\theta)$ is given by the information inequality

$$\text{Var}(T) \geq \frac{[\tau'(\theta)]^2}{-nE\left[\frac{d^2}{d\theta^2} \ln f(X|\theta)\right]}.$$

Now complete the following tasks:

(a) Specify the joint *pdf* of $X_1, \cdots, X_n$.

(b) Find the score function $U(\theta) = \frac{d}{d\theta} \ln L(\theta)$ and the MLE of θ based on the sample.

(c) Let $\phi = \theta^{-1}$. Find the MLE of ϕ.

(d) Show that $E[\ln X] = -\theta^{-1}$. Then use this result to show that the MLE of ϕ, denoted as $\hat{\phi}$, is unbiased for ϕ.

(e) Show that $E[(\ln X)^2] = 2\theta^{-2}$. Then show that the MLE $\hat{\phi}$ is a UMVUE (uniformly minimum variance unbiased estimator) of $\phi = \theta^{-1}$.

[1 + 2 + 2 + 3 + 4 = 12 marks]

2. Consider a GLM (generalised linear model) for random samples $Y_1, \cdots, Y_n$ which are to be observed from binomial distributions. Here each Y_i has the *pmf*

$$f(y_i|p_i) = \binom{m_i}{y_i} p_i^{y_i}(1-p_i)^{m_i-y_i}, \quad y_i = 0, 1, \cdots, m_i; \; 0 < p_i < 1; \quad i = 1, \cdots, n;$$

where m_i is also to be observed, $E(Y_i) = m_i p_i$ and $\text{Var}(Y_i) = m_i p_i(1-p_i)$. Suppose the systematic component of the GLM is $\eta_i = \alpha + \beta x_i$, with (α, β) being the parameters of interest and x_i's being the covariate observations; and the link is $\eta_i = \ln \frac{p_i}{1-p_i}$, $i = 1, \cdots, n$. Note the following results can be derived from the model setup and be quoted without proof:

(i) $p_i = \frac{e^{\eta_i}}{1+e^{\eta_i}} = \frac{e^{\alpha+\beta x_i}}{1+e^{\alpha+\beta x_i}}$.

(ii) $\frac{\partial p_i}{\partial \alpha} = \frac{e^{\alpha+\beta x_i}}{(1+e^{\alpha+\beta x_i})^2} = p_i(1-p_i)$.

(iii) $\frac{\partial p_i}{\partial \beta} = \frac{x_i e^{\alpha+\beta x_i}}{(1+e^{\alpha+\beta x_i})^2} = x_i p_i(1-p_i)$.

Now complete the following tasks:

(a) Find the log-likelihood function $\ell(\alpha, \beta) = \ln L(\alpha, \beta)$.

(b) Find $\frac{\partial \ln L}{\partial \alpha}$ and $\frac{\partial \ln L}{\partial \beta}$ which specify the score function $U(\alpha, \beta)$.

(c) Find $\frac{\partial^2 \ln L}{\partial \alpha^2}$, $\frac{\partial^2 \ln L}{\partial \beta^2}$ and $\frac{\partial^2 \ln L}{\partial \alpha \partial \beta}$. Then list the Fisher information matrix $I(\alpha, \beta)$.

(d) Given that $(\alpha^{(j)}, \beta^{(j)})$ is the current estimate in the process of finding the MLE of (α, β), we can update it to $(\alpha^{(j+1)}, \beta^{(j+1)})$ by the Newton-Raphson or Fisher scoring procedure. Specify the formula used in this update.

[2 + 2 + 5 + 3 = 12 marks]

3. We have random samples $(Y_1, X_1), \cdots, (Y_n, X_n)$ from the model $Y_i = X_i + \varepsilon_i$, $i = 1, \cdots, n$, where each X_i has the pdf $f(x_i|\lambda) = \lambda e^{-\lambda x_i}$, $x_i > 0$, $\lambda > 0$; each ε_i has the pdf $g(\varepsilon_i) = e^{-\varepsilon_i}$, $\varepsilon_i > 0$; and all X_i's and ε_i's are mutually independent of each other.

   Suppose we have observed $y_1, \cdots, y_n$ for $Y_1, \cdots, Y_n$ only. We want to use an EM algorithm to find the MLE (maximum likelihood estimate) of λ.

   (a) We may treat $(Y_1, X_1), \cdots, (Y_n, X_n)$ as the complete data. Show that the joint pdf for each (Y_i, X_i) pair is

   $$f(y_i, x_i|\lambda) = \lambda e^{-(\lambda-1)x_i - y_i}, \quad y_i > x_i > 0.$$

   Then find the complete-data log-likelihood function $\ell(\lambda) = \ln L(\lambda)$.

   (b) Show that the conditional pdf of X_i given $Y_i = y_i$ is

   $$f(x_i|y_i, \lambda) = \frac{(\lambda-1)e^{-(\lambda-1)x_i}}{1 - e^{-(\lambda-1)y_i}}, \quad 0 < x_i < y_i.$$

   Then show that

   $$E(X_i|Y_i = y_i, \lambda) = \frac{1}{\lambda - 1} + \frac{y_i e^{-(\lambda-1)y_i}}{e^{-(\lambda-1)y_i} - 1}.$$

   (c) In the E-step, we calculate

   $$Q(\lambda, \lambda^{(k)}) = E[\ln L(\lambda) \mid Y_1 = y_1, \cdots, Y_n = y_n, \lambda^{(k)}]$$

   where $\lambda^{(k)}$ is the current estimate of λ. Show that

   $$Q(\lambda, \lambda^{(k)}) = n \ln \lambda - (\lambda - 1) \sum_{i=1}^{n} \left[\frac{1}{\lambda^{(k)} - 1} + \frac{y_i e^{-(\lambda^{(k)}-1)y_i}}{e^{-(\lambda^{(k)}-1)y_i} - 1} \right] - \sum_{i=1}^{n} y_i.$$

   (d) In the M-step, we maximise $Q(\lambda, \lambda^{(k)})$ with respect to λ to find an update $\lambda^{(k+1)}$ from $\lambda^{(k)}$. Show that

   $$\lambda^{(k+1)} = n \left(\sum_{i=1}^{n} \left[\frac{1}{\lambda^{(k)} - 1} + \frac{y_i e^{-(\lambda^{(k)}-1)y_i}}{e^{-(\lambda^{(k)}-1)y_i} - 1} \right] \right)^{-1}.$$

   [3 + 5 + 3 + 1 = 12 marks]

4. Let $X_1, \cdots, X_n$ be random samples from geometric distribution Geo(θ). Thus the *pmf* of X_i given θ is

   $$f(x_i|\theta) = \theta(1-\theta)^{x_i - 1}, \quad x_i = 1, 2, \cdots; \quad i = 1, \cdots, n.$$

   Suppose θ has a Beta(a, b) prior distribution with the *prior pdf* being

   $$p(\theta|a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1}(1-\theta)^{b-1}; \quad a > 0, \ b > 0 \text{ being given}; \quad \text{also } E(\theta) = \tfrac{a}{a+b}.$$

   (a) Find the posterior pdf of θ given $X_1, \cdots, X_n$.

   (b) Find the Bayes estimator of θ when the squared loss function is used.

   (c) Find the Bayes estimator of θ^{-1} when the squared loss function is used.

   (d) Find a 95% Bayes credible interval for θ.

   [4 + 2 + 3 + 1 = 10 marks]

5. A random number X is to be generated by the following acceptance-rejection (A-R) sampling algorithm:

   1° Generate $U = u$ from Unif$(0,1)$ and $V = v$ from Unif$(0,1)$ independently.

   2° If $u + v > 1$, deliver $X = -\ln(u)$; otherwise, go to 1°.

   The *cdf* (cumulative distribution function) of X can be found using the following result:

   $$P(X \le x) = P(-\ln(U) \le x \mid U + V > 1), \quad \text{for a given } x.$$

   (a) Show that the pdf of X is $f(x) = 2e^{-2x}, \quad x > 0.$

   (b) Let N be the number of times steps 1° and 2° need be repeated before an X value is generated. What is the distribution of N and what is the mean $E(N)$?

   [5 + 3 = 8 marks]

6. Consider a random sample $X_1, \cdots, X_n$ from Gamma$(\theta, 5)$ distribution which has the pdf

   $$f(x|\theta) = \frac{1}{24\theta^5} x^4 e^{-\frac{x}{\theta}}; \quad x > 0, \ \theta > 0.$$

   It can be shown that θ is a scale parameter and $\hat{\theta} = \frac{1}{5}\bar{X}_n = \frac{1}{5n}\sum_{i=1}^{n} X_i$ is the MLE of θ. It can also be shown that $Q = 2n\theta^{-1}\bar{X}_n \overset{d}{=} \chi^2(10n)$ and Q is a pivotal quantity.

   (a) Explain what is called a pivotal quantity.

   (b) Derive a 95% C.I. (confidence interval) formula for θ based on Q.

   (c) Based on a sample of 50 observations from Gamma$(\theta, 5)$, 1999 nonparametric bootstrap samples are generated to estimate se$(\hat{\theta})$ and construct C.I. for θ. The following R output was obtained where x stores the 50 generated Gamma$(\theta, 5)$ observations:

```
> library(boot)
> scale.bt=function(x,i){scale=mean(x[i])/5
        n=length(x)
        v.scale=(n-1)*var(x[i])/(25*n^2)
        c(scale,v.scale)}
> set.seed(12345)
> bootQ6=boot(data=x,statistic=scale.bt,R=1999)
> bootQ6

ORDINARY NONPARAMETRIC BOOTSTRAP

Call:
boot(data = x, statistic = scale.bt, R = 1999)

Bootstrap Statistics :
     original       bias    std. error
t1* 8.1700864 -0.01199792  0.58243413
t2* 0.3471393 -0.00750799  0.06640543

> boot.ci(bootQ6)

BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
Based on 1999 bootstrap replicates

CALL :
boot.ci(boot.out = bootQ6)

Intervals :
Level      Normal              Basic             Studentized
95%   ( 7.041,  9.324 )   ( 7.039,  9.320 )   ( 7.092,  9.498 )

Level     Percentile            BCa
95%   ( 7.020,  9.301 )   ( 7.056,  9.353 )
Calculations and Intervals on Original Scale
```

```
> qchisq(0.975,500); qchisq(0.025,500)
[1] 563.8515
[1] 439.936
```

i. From the R output find the value of $\hat{\theta}$, the bootstrap estimate of se($\hat{\theta}$), and the 95% Studentized C.I. of θ. Explain how the 95% Studentized C.I. of θ is calculated.

ii. Calculate the 95% C.I. for θ using the formula obtained in (b) and the χ^2 percentile values in the R output.

[1 + 3 + 4 + 1 = 9 marks]

7. **Neyman-Pearson Lemma:** Suppose random variables $X_1, \cdots, X_n$ have the joint pdf $f(\mathbf{x}_n|\theta)$ with $\mathbf{x}_n = (x_1, \cdots, x_n)^T$. Let $\lambda(\mathbf{x}_n) = \dfrac{f(\mathbf{x}_n|\theta_1)}{f(\mathbf{x}_n|\theta_0)}$ be the likelihood ratio test statistic, and $C^* = \{\mathbf{x}_n : \lambda(\mathbf{x}_n) \geq k\}$ be a critical region for testing

$$H_0\colon\ \theta = \theta_0 \quad \text{vs.} \quad H_a\colon\ \theta = \theta_1,$$

where k is such that $P(\mathbf{X}_n \in C^* \mid \theta_0) = \alpha$. Then C^* is the most powerful critical region of size α for the test. Namely, for any critical region C such that $P(\mathbf{X}_n \in C|\theta_0) \leq \alpha$,

$$\pi_{C^*}(\theta_1) = P(\mathbf{X}_n \in C^* \mid \theta_1) \geq P(\mathbf{X}_n \in C \mid \theta_1) = \pi_C(\theta_1).$$

Question: Consider i.i.d. random samples $X_1, \cdots, X_n$ of X that has the *pdf*

$$f(x|\theta) = \frac{2}{\theta^2} x e^{-\left(\frac{x}{\theta}\right)^2};\ \ x > 0,\ \theta > 0.$$

It can be shown that $X_i^2 \overset{d}{=} \text{Exp}(\theta^2)$, $i = 1, \cdots, n$; and $\hat{\theta} = \sqrt{n^{-1}\sum_{i=1}^n X_i^2}$ is the MLE of θ.

(a) Determine an MP (most powerful) critical region for testing $H_0\colon\ \theta = \theta_0$ vs. $H_a\colon\ \theta = \theta_1$, where $\theta_1 > \theta_0$, at significance level α. Simplify the critical region as much as you can.

(b) Explain whether the critical region found in (a) is UMP (uniformly most powerful) of size α for testing $H_0\colon\ \theta \leq \theta_0$ vs. $H_a\colon\ \theta > \theta_0$.

(c) Derive a GLR (generalised likelihood ratio) test for testing $H_0\colon\ \theta = \theta_0$ vs. $H_a\colon\ \theta \neq \theta_0$. Simplify the derived critical region as much as you can.

[3 + 3 + 3 = 9 marks]

8. Twenty members of a focus group were asked to view a political debate between two candidates A and B who run for the president. They were asked before the debate whom they would vote as the president; and were asked again after the debate whom they would vote. Data on whom they would vote before and after the debate are shown in the following table.

	Vote A after the debate	Vote B after the debate
Vote A before the debate	3	9
Vote B before the debate	2	6

The question of interest is whether the probability of voting A before the debate is different from the probability of voting A after the debate. This can be answered by using McNemar's method to test $H_0\colon\ p_1 = p_2$ vs. $H_a\colon\ p_1 \neq p_2$, where p_1 is the probability of a member voting A before the debate and p_2 is that after the debate.

(a) Let Y_{12} denote the top-right observation in the table. What is the distribution of Y_{12} when H_0 is true? Find the exact p-value of the test using McNemar's binomial test statistic.

(b) The following is the R output from the McNemar's χ^2 test. Explain how the test statistic value $\chi^2 = 3.2727$ is obtained.

```
> x=matrix(c(3,2,9,6),2,2)
> mcnemar.test(x, correct=T)

        McNemar's Chi-squared test with continuity correction

data:  x
McNemar's chi-squared = 3.2727, df = 1, p-value = 0.07044
```

[4 + 2 = 6 marks]

9. The following table refers to a sample of subjects randomly selected for an Italian study on the relation between income and whether one possesses a travel credit card. At each level of annual income in millions of lira (x), the table indicates the number of subjects sampled (n) and the number of them possessing at least one travel credit card (y).

y	0	2	1	0	1	1	0	0	0	2	0	6	3	0	0	1
n	1	5	9	8	7	3	5	2	1	10	1	6	5	1	1	1
x	24	28	30	32	34	38	40	42	48	50	59	65	70	80	94	130

A logistic model $\ln \frac{\theta}{1-\theta} = \beta_0 + \beta_1 x$, where θ is the probability of a subject possessing at least one travel credit card, is fitted to the data. The relevant results are provided in the following R output.

```
> card.2 <- glm(y/n ~ x,family=binomial,weight = n,data=card.dat)
> anova(card.2, test="Chi")

Analysis of Deviance Table

Model: binomial, link: logit

Response: y/n

Terms added sequentially (first to last)


     Df Deviance Resid. Df Resid. Dev P(>|Chi|)
NULL                    15     35.999
x     1   10.597        14     25.402 0.001133

> 1-pchisq(25.402,14)

[1] 0.03080005

> summary(card.2)$coef

              Estimate Std. Error   z value     Pr(>|z|)
(Intercept) -3.54362777 0.95041274 -3.728515 0.0001926118
x            0.05220598 0.01841121  2.835554 0.0045746330

> summary(card.2)$cov.unscaled

            (Intercept)             x
(Intercept)  0.90328438 -0.0165478144
x           -0.01654781  0.0003389727
```

(a) Use both Wald test and the (generalised) likelihood ratio test to assess the significance of annual income level x on affecting the probability of possessing at least one travel credit card based on model `card.2`. Report the test statistic value and the p-value for each test.

(b) Does the model $\ln \frac{\theta}{1-\theta} = \beta_0 + \beta_1 x$ provide an adequate fit to the data? Why or why not?

(c) Regardless of the result of (b), use the model $\ln \frac{\theta}{1-\theta} = \beta_0 + \beta_1 x$ to estimate the odds ratio of possessing at least one travel credit card when a subject's annual income increases by 10 millions of lira. Also find an approximate 95% C.I. for this odds ratio.

(d) Use the model $\ln \frac{\theta}{1-\theta} = \beta_0 + \beta_1 x$ to estimate the probability of possessing at least one travel credit card if a subject has an annual income of 100 millions of lira. Also find an approximate 95% C.I. for this probability.

[2 + 2 + 3 + 4 = 11 marks]

10. Data on the presence or absence of a certain insect on plants were taken at three locations. The study used two types of plants: those that are not resistant to the insect and those that were bred to be resistant. The data are summarized in the following $2 \times 2 \times 3$ table.

Location	Type of Plants	Insect present	Insect absent
Location 1	Resistant	4	16
	Not Resistant	9	11
Location 2	Resistant	1	18
	Not Resistant	6	14
Location 3	Resistant	7	11
	Not Resistant	10	9

(a) A Mantel-Haenszel test is performed returning the following R output. Explain what this test is testing for and what the conclusion is at the significance level 0.05.

```
> q10=array(0,c(2,2,3))
> q10[,,1]=c(4,9,16,11)
> q10[,,2]=c(1,6,18,14)
> q10[,,3]=c(7, 10, 11, 9)
> mantelhaen.test(q10,correct=T)

        Mantel-Haenszel chi-squared test with continuity correction

data:  q10
Mantel-Haenszel X-squared = 5.3055, df = 1, p-value = 0.02126
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval:
 0.1436797 0.7971515
sample estimates:
common odds ratio
        0.3384295
```

(b) The data in the table are rearranged so that a log-linear model `mod1` can be fitted, with the results being given in the following R output. Give the mathematical formula (of the form $\ln(\lambda) = \cdots$) for model `mod1`.

```
> q10A.dat=data.frame(freq=c(4, 9, 16, 11,  1,  6, 18, 14,  7, 10, 11,  9))
> q10A.dat$Location=factor(rep(c("Loc1","Loc2", "Loc3"), each=4))
> q10A.dat$Type=factor(rep(c("Resist","NotResist"), times=6))
> q10A.dat$Presence=factor(rep(c("Present","Absent"),times=3, each=2))
> q10A.dat
   freq Location      Type Presence
1     4     Loc1    Resist  Present
2     9     Loc1 NotResist  Present
3    16     Loc1    Resist   Absent
4    11     Loc1 NotResist   Absent
5     1     Loc2    Resist  Present
6     6     Loc2 NotResist  Present
7    18     Loc2    Resist   Absent
```

```
8    14     Loc2 NotResist   Absent
9     7     Loc3    Resist  Present
10   10     Loc3 NotResist  Present
11   11     Loc3    Resist   Absent
12    9     Loc3 NotResist   Absent
```

```
> mod1=glm(freq~Location+Type+Presence+Location:Type+Location:Presence+Type:Presence,
 +  family=poisson, data=q10A.dat)
> anova(mod1, test="Chisq")
```

```
Analysis of Deviance Table

Model: poisson, link: log

Response: freq

Terms added sequentially (first to last)

                 Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                                11    30.8286
Location          2   0.1212         9    30.7074   0.94119
Type              1   0.0345         8    30.6729   0.85268
Presence          1  15.5579         7    15.1150 8.002e-05
Location:Type     2   0.0182         5    15.0968   0.99095
Location:Presence 2   7.0484         3     8.0484   0.02948
Type:Presence     1   6.5699         2     1.4785   0.01037
---
```

```
> 1-pchisq(1.4785,2)

0.4774719

> 1-pchisq(8.0484,3)

[1] 0.04502194
```

(c) What assumption is made in order for the Mantel-Haenszel test in (a) to be valid, in regard to the association between the presence/absence of insect and the plant type? Test this assumption using the R output in (b) and draw the conclusion at the significance level 0.05.

(d) Test based on `mod1` the significance of the interaction effect `Type:Presence`, eliminating the effects of all other terms in the model. Comment on the implication of your result and how that is related to the result of (a).

(e) Test the adequacy of model `Location+Type+Presence+Location:Type+Location:Presence`, at significance level 0.05, using the R output in (b). Comment on the implication of your result and how that is related to the results of (a) and (d).

[2 + 2 + 3 + 2 + 2 = 11 marks]

Total marks = 100

END OF EXAMINATION